

Vincent Wu

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Education

North Carolina State University

Master of Science (Accelerated Program), Electrical Engineering

Expected May 2028

Bachelor of Science, Electrical Engineering – GPA: 3.9

Expected May 2027

- Minor, Chinese Studies

Experience

Electronics Lead – Pack Motorsports Baja SAE (Sept 2024 - Present)

- Designed and integrated custom Hall effect sensors for suspension displacement, wheel speed, and piezo-based engine RPM sensing using KiCAD and OnShape
- Implemented a Python backend to collect CAN bus data and log to MF4 files
- Developing a dashboard using Vue.js for real-time data visualization
- Manage multiple concurrent projects and coordinate with other subsystem teams

Mechatronics Engineer Intern – Volvo Group (May 2025 - Aug 2025)

- Designed an automated system to reduce driver efforts during trailer setup
- Implemented real-time control logic for the automated system using Simulink and Speedgoat computers
- Created custom UI for the system using Next.js and Flask

Mechatronics Engineer Intern – Volvo Group (May 2024 - Aug 2024)

- Developed traction features involving airbag suspension
- Implemented simulation of differential lock into hardware-in-the-loop testing
- Analyzed HIL development workflows and proposed workflow improvements that increased simulation efficiency

Undergraduate Research Assistant – AERPAW (Jan 2024 - May 2024)

- Assembled electrical systems for a rover-type wireless communication testbed
- Researched more reliable IMU sensing for drone testbeds

Lead Instructor – Queen City Robotics Alliance (Jun 2023 - Jul 2023)

- Led the instruction of multiple camps to teach 4th through 8th graders how to program in Python through the use of drones
- Prepared teaching materials and custom tools to enhance the students' learning

Engineering Captain – YETI Robotics (Dec 2022 - Jul 2023)

- Coordinated the cooperation between mechanical, electrical, and software subteams to develop a World Championship level robot
- Advised subteam managers on solutions and priorities
- Assessed and supported the needs of individual team members and the team as a whole

Licenses & Certifications

Certified SolidWorks Professional – Certificate ID: C-Q8JUKW69A7

Skills

Programming

C/C++ | Python | Java | Go | MATLAB | Vue.js | Next.js | Flask | Git

Hardware & Design

KiCAD | SolidWorks | OnShape | HIL

Embedded & Controls

Simulink | Real-time computers | Arduino | Espressif

Honors & Activities

University Theatre - Scene shop, sound board operator, spotlight operator

Sept 2023 - Present

IEEE HKN Beta Eta Chapter

Nov 2024 - Present